

Winter Storm Fern Situation Report

January 2026

Formatted based on LA Fire [Situation Report](#)

Summary

A major American winter storm, often referred to as Winter Storm Fern, started on Friday, January 23rd 2026 and continued through January 26th 2026. The storm brought heavy snow and freezing rain to several U.S. states, ranging from the southern plains to the Eastern United States. Several states experienced network outages and extremely cold temperatures. For example, emergency declarations were issued in Arkansas, Georgia, Indiana, Kentucky, Louisiana, Maryland, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, and West Virginia (Congressional Research Service 2026). Power outages due to downed trees and ice occurred in Southern States, such as Texas, Louisiana, Mississippi, and Tennessee. States from New England to New Mexico experienced significant snowfall, and sleet occurred in the Mid-Atlantic and Northeast. As of January 29th, there were up to 115 fatalities across 20 states after this winter storm, and approximately 2.5 million customers experienced power outages across the United states (Kothari 2026). Verisk, who are catastrophe risk modelling specialists, estimated a total of US \$4 billion in industry losses with 14 states, ranging from Massachusetts to Texas, that could each exceed \$50 million in insured losses (Evans 2026).

The following visualizations are developed based on Meta AI for Good's data through Facebook with a focus on Tennessee and Kentucky.. The data covers network coverage and population density from January 27 to February 10, with a focus on Tennessee and Kentucky and daily updates during the collection period.

How the Data Are Constructed

The Facebook data used in this report comes from Meta's Data for Good crisis datasets. In plain terms, these datasets use activity from Facebook users or Facebook business Pages to estimate how people, population density, network coverage, and business activity change

during a crisis. The data does not represent everyone in the affected area. For population and movement datasets, it only represents Facebook app users who have Location Services enabled. For business activity, it represents qualifying Facebook business Pages with enough activity to be included while preserving privacy.

The data is constructed by comparing what is observed during the crisis period to what was typical before the crisis. For example, population density during the storm is compared to a pre-crisis baseline. Movement during the storm is compared to normal movement patterns between the same places before the event. Business activity is compared to normal posting behavior from business pages before the crisis. Because the data is relative to a baseline, the main signal is not the raw count itself, but whether a place is above or below its expected level.

Geographically, the data is reported in two main ways: Bing tiles and administrative regions. Bing tiles divide space into small grid cells. At Bing tile level 14, each tile is roughly 2.4 kilometers on a side near the equator, or about the size of several city blocks. Administrative regions instead use boundaries such as counties, states, or other official geographic areas. These are two different approximations of the same general region and same time window, but they can show different rates of population density change (Figure 1).

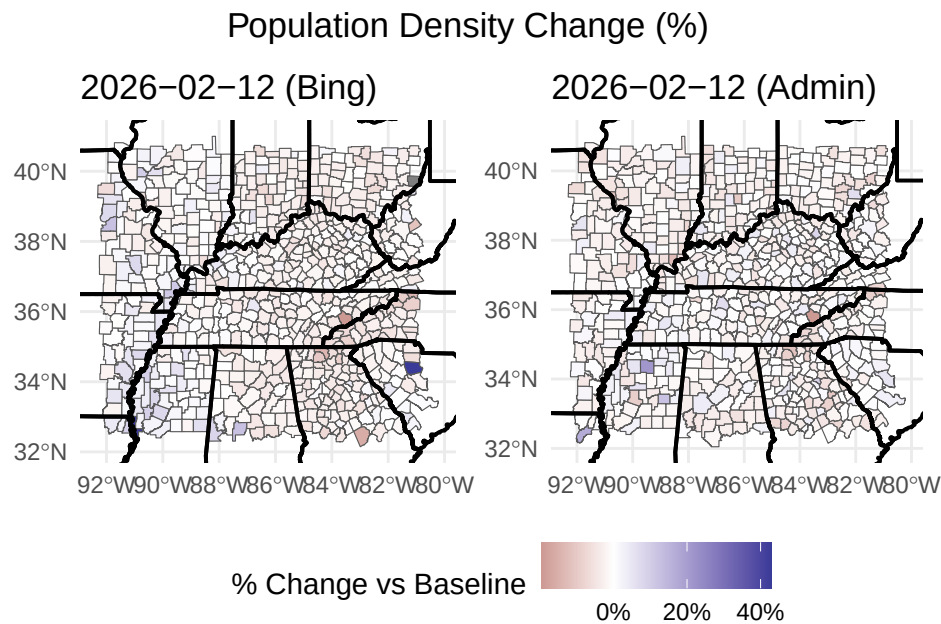


Figure 1: Changes in population density vary when represented by data from bing tiles or by data from administrative regions, even in the same region and time window.

Takeaways (Figure 1)

- Bing tile results and administrative-region results should not be treated as interchangeable
- The strongest changes appear concentrated in a small number of counties rather than spread evenly across the region
- For example, on February 12th, the population density change for Issaquena, Mississippi is 59.7% based on Bing tiles and only -2% based on administrative regions.

Temporally, the population and movement datasets use fixed 8-hour windows. These windows begin at 00:00, 08:00, and 16:00 Pacific Time. This means that the time periods do not automatically adjust to local time zones. For Tennessee and Kentucky, the reported time windows are still based on Pacific Time, which should be kept in mind when interpreting daily patterns. Business activity is reported daily, based on the date value in the dataset. Some Meta crisis datasets are provided directly as Bing tile data, while others are provided as administrative-region data. When the data is provided as Bing tiles, each observation is tied to a grid cell with a latitude and longitude. When we add county information to those data, we are not changing the original measurement. Instead, we are assigning each tile or point to the county that contains it. This allows the tile-level data to be summarized at the county level.

For example, a Bing tile located inside Davidson County, Tennessee would be assigned the Davidson County name and GEOID. If many Bing tiles fall within Davidson County, their values can then be averaged, summed, or otherwise summarized to produce a county-level measure. This is useful for reporting, but it is not identical to receiving an administrative-region dataset directly from Meta. The administrative-region dataset is already aggregated within county or state boundaries by Meta, while the Bing tile version requires us to overlay the tile locations onto county boundaries.

Facebook Data Definitions

- **Baseline Ratio:** Ratio of people using the Facebook app who have turned on the Location Services device setting on their mobile device that are present in a specific tile during the baseline period over users in all tiles in the boundary box defining the crisis
- **Crisis Ratio:** Ratio of people using the Facebook app who have turned on the Location Services device setting on their mobile device that are present in a specific tile during the 8-hour window over users in all tiles in the boundary box defining the crisis
- **Population Change Density (%):** Percent change in Facebook Location Services-enabled user population counts per tile, calculated by dividing the difference by the baseline (plus a small value, usually 1)
- **Time Windows:** aggregated over 3 fixed 8-hour time windows, beginning at 00:00, 08:00 and 16:00, that represent changes in population density

- **Bing Tiles:** equivalent to roughly 2.4 kilometers on a side near the equator or the size of 6 x 6 city blocks
- **Administrative Regions:** the polygon boundaries for administrative regions derived from the Database of Global Administrative Areas, also known as GADM.

Outside Data Added

In addition to the Meta crisis datasets, county-level demographic data was added from the American Community Survey in 2022 (U.S. Census Bureau 2025). This demographic data was incorporated with the use of the `tidycensus` package (Walker and Herman 2025). This outside data provides context about the counties affected by the storm, including total population, median household income, poverty rate, share of residents age 65 and older, and share of households without vehicle access. These variables help interpret which communities may be more vulnerable during a crisis.

The demographic data is joined using `county_geoid`, a standardized county identifier. This is important because county names alone are not unique across states. For example, many states have counties with the same name, but each county has a unique GEOID.

Aggregation Choices

For county-level maps, the data is summarized by county and date. When multiple observations fall within the same county on the same date, the report uses the average percent change to represent the county. This choice is appropriate for percent-based indicators because summing percentages would not be meaningful.

For population count variables, such as baseline population or crisis population, summing can be used when the goal is to estimate total observed users across tiles or regions. For percent change, z-scores, or ratios, averaging is more appropriate because these variables already describe relative change.

For movement data, origin and destination counties are treated separately. Each row represents movement from a starting location to an ending location, so the analysis keeps both `start_county_geoid` and `end_county_geoid`. This avoids incorrectly collapsing movement into a single county and preserves the direction of movement.

Limitations

These datasets should be interpreted as indicators of crisis-related change, not exact measures of the full population. The population and movement data only include Facebook users with Location Services enabled, so the sample may not represent all residents equally. Areas with fewer Facebook users or lower Location Services usage may be less reliable.

Privacy protections also affect interpretation. Small counts may be suppressed or removed, which can especially affect rural areas. For movement data, this means it is not reliable to calculate exact net flows by simply adding inbound and outbound movement, because some small movement vectors are excluded.

The baseline period is limited to the pre-crisis period available to Meta. This means the analysis does not have a long historical record of pre-crisis behavior. As a result, seasonal changes, holidays, or unusual pre-crisis conditions may affect the baseline. The data is useful for rapid situational awareness, but it should not be interpreted as a complete long-term population or economic dataset.

Finally, the data is time-limited. The documentation notes that some datasets are accessible for 90 days after the final update day and then discarded, which limits longer-term reproducibility and longitudinal analysis.

Movement

% Population Density Change

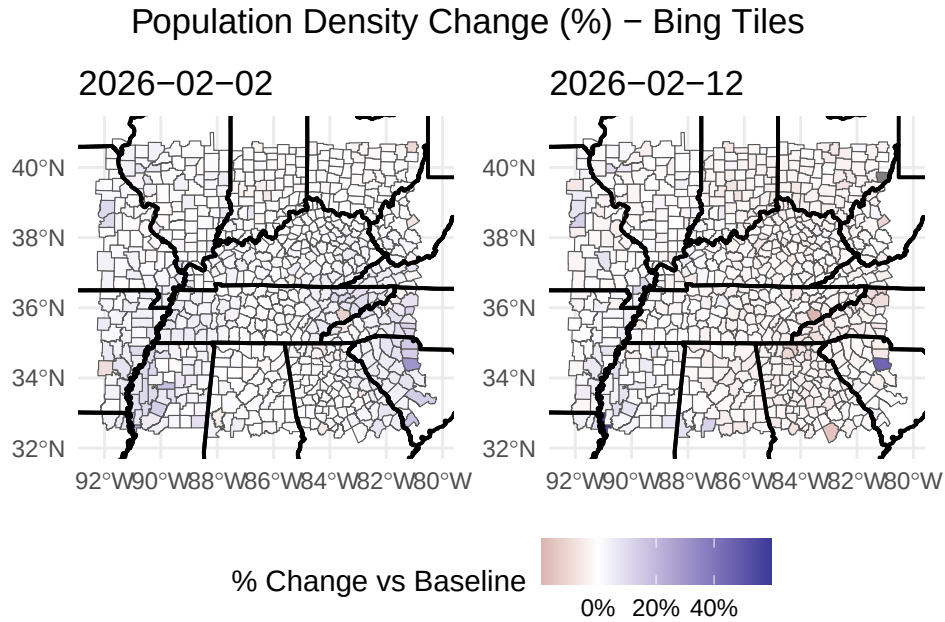


Figure 2: When showing population density change with Bing tiles, there is little negative changes at these two dates.

Takeaways (Figure 2)

- The overall spatial pattern appears fairly stable between the two dates
- There is little visible negative population-density change in the Bing tile version
- On February 2nd, the counties with the most increased population density are (Table 2):
 - Fairfield, South Carolina (30.5%)
 - Issaquena, Mississippi (25.5%)
 - Grenada, Mississippi (17.9%)
- On February 2nd, the counties with the most decreased population density are (Table 3):
 - Sevier, Tennessee (-12.5%)
 - Carroll, Ohio (-9.6%)

- Jefferson, Arkansas (-8.9%)
- On February 12th, the counties with the most increased population density are (Table 4):
 - Issaquena, Mississippi (59.7%)
 - Fairfield, South Carolina (45.2%)
 - Osage, Missouri (12.6%)
 - Perry, Alabama (12.6%)
- On February 12th, the counties with the most decreased population density are (Table 5):
 - Sevier, Tennessee (-19.6%)
 - Laurens, Georgia (-15.6%)
 - Fannin, Georgia (-13.8%)
 - Alleghany, North Carolina (-13.8%)

Time Series of Movement

The counties are grouped by average absolute population-density change over the recording period.¹

¹The percent changes are summarized over the full recording period for each county. For each county, the absolute value of the percent change is taken at each time point, and those absolute changes are averaged across the 14-day period. This creates a single measure of how much each county deviated from baseline, regardless of whether the change was positive or negative. Counties are then compared using this average absolute change to identify the highest-change, lowest-change, and average-change counties. The county-level change measure is defined as $\text{AvgAbsChange}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} |p_{it}|$, where p_{it} is the percent change for county i at time t , and T_i is the number of time periods observed for that county.

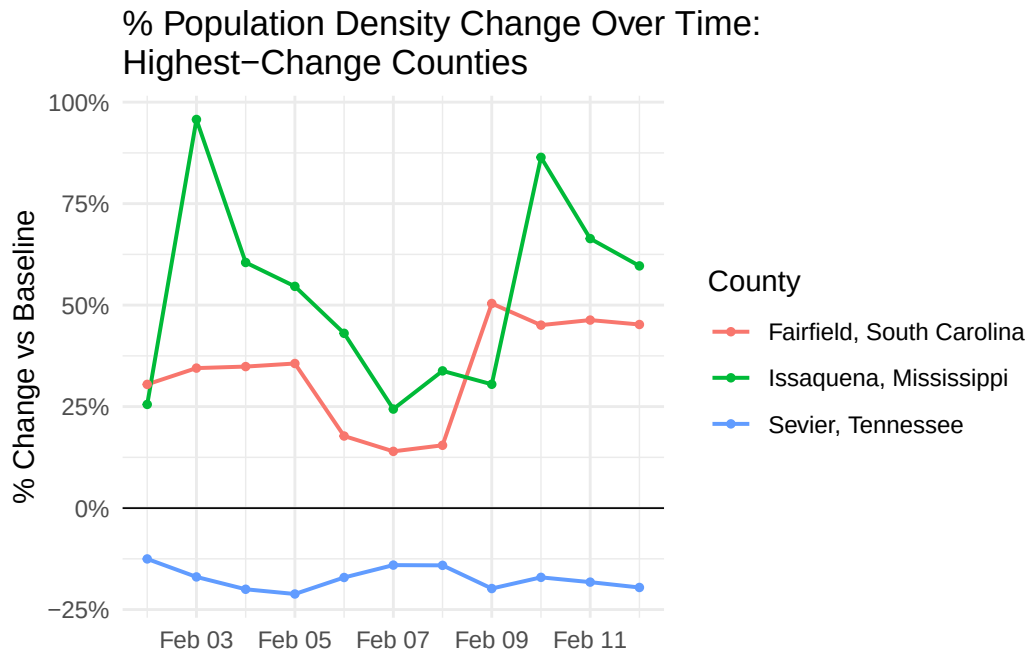


Figure 3: The three counties with the highest change have positive change or negative change during this entire time period.

Takeaways (Figure 3)

- Fairfield and Issaquena show the largest positive population-density changes over the time period.
- Issaquena rises sharply near the end of the period, reaching close to 100% change.
- Sevier is included among the highest-change counties because of large absolute change, but its trend is negative rather than positive.
- The figure shows that “highest change” includes both large increases and large decreases when counties are selected using absolute change.
- The time series makes clear that the largest changes are persistent over multiple days, not just one-day spikes.
- Both Fairfield and Issaquena almost constantly have population density change over 25%.

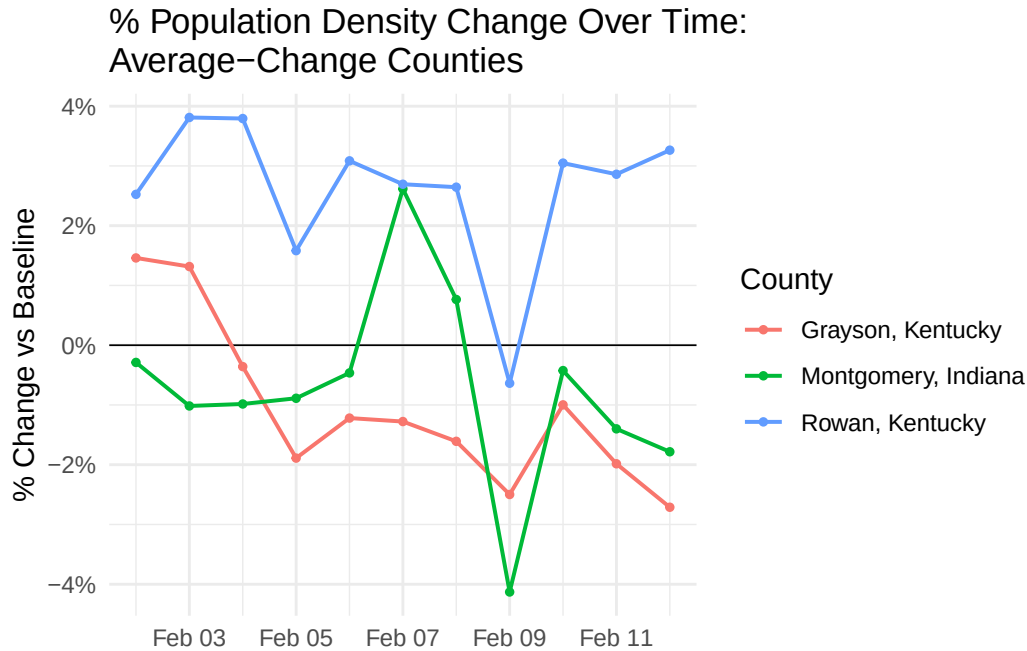


Figure 4: The three counties with average change have more variation in population density change with both positive and negative changes.

Takeaways (Figure 4)

- These counties provide a useful comparison group for interpreting the more extreme counties.
- In comparison to the high-change counties, these three counties have their percentage population density change constrained within $[-4\%, 4\%]$, indicating much lower rates of population density change.
- All counties have a drop in population density change at February 9th but with varying magnitudes.
- The Grayson county has a somewhat monotonic decline with people initially coming in and then steadily moving out.
- The Rowan county has constantly positive population density change, indicating incoming population, with a sharp drop on February 9th.
- The Montgomery has constantly negative population density, indicating leaving populations, with a sharp spike at February 7th and a sharp drop at February 9th.

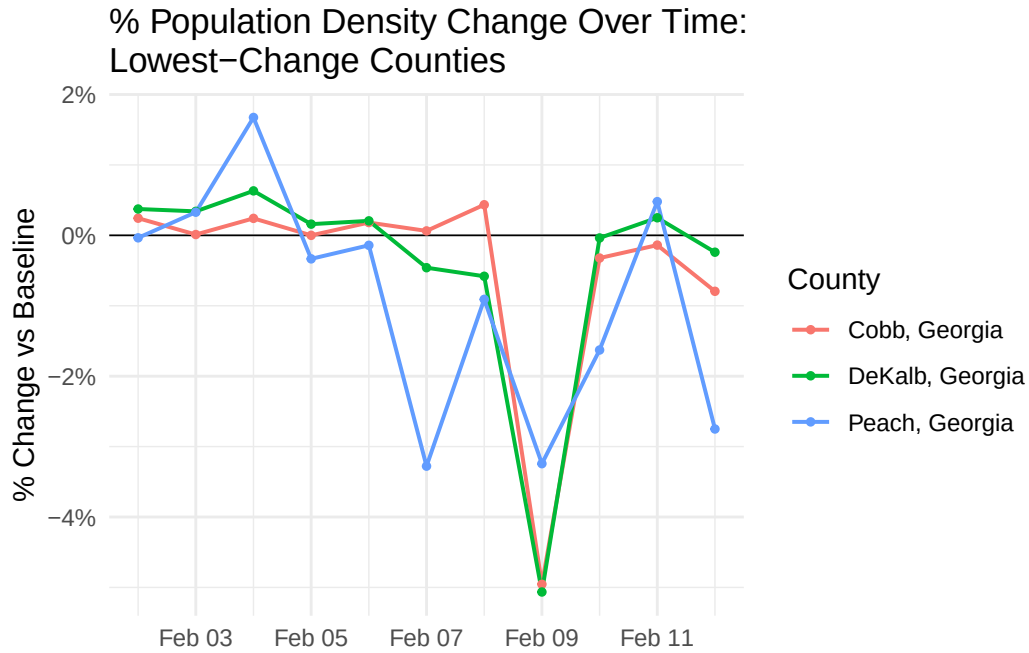


Figure 5: The three counties with the lowest change have a notable drop into extreme negative population density change on February 9th.

Takeaways (Figure 5)

- These three counties follow a similar pattern with relatively stable populations around zero and a sharp drop at February 9th.
- Cobb and DeKalb follow this pattern very closely and very similarly.
- The Peach county has more variation with spikes at February 7th, February 9th and February 11th.
- These counties never have strong positive population density change, implying that populations may have either entered before this time period or left after this time period, because the constant negative change indicates people leaving these counties.

Bottom 20 county-to-county crisis movement flows

Source: Bing Tiles

start_county_name	end_county_name	total_crisis_movement	avg_z_score
Carroll	Fulton	0	NaN
Fulton	Sevier	0	NaN
Lafayette	Panola	0	NaN
Abbeville	Anderson	10	NaN
Adair	Russell	10	NaN
Arkansas	Prairie	10	NaN
Auglaize	Shelby	10	NaN
Barren	Warren	10	NaN
Bedford	Coffee	10	NaN
Blount	Sevier	10	NaN
Bolivar	Washington	10	NaN
Boone	Gallatin	10	NaN
Boyd	Cabell	10	NaN
Bradley	Polk	10	NaN
Brown	Morgan	10	NaN
Buncombe	McDowell	10	NaN
Burke	Richmond	10	NaN
Calloway	McCracken	10	NaN
Calloway	Vigo	10	NaN
Campbell	Anderson	10	NaN

Table 1: The bottom 20 county-to-county crisis movement flows are all at zero or ten, likely due to data-collection thresholds for privacy concerns.

Places

Takeaways (Table 1)

- The table is more useful as a data-quality warning than as a substantive movement analysis.
- Because movement vectors with small counts can be suppressed, these low-flow values should not be interpreted as exact zero movement.

Counties with Most Increased Population Density on 2026-02-02

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Fairfield	South Carolina	30.5	20,550	\$47,885	15.6%	3.2%	5.9%
Issaquena	Mississippi	25.5	928	\$31,429	16.2%	2.7%	14.0%
Grenada	Mississippi	17.9	21,217	\$48,804	24.6%	2.3%	7.9%
Hampton	South Carolina	16.6	18,254	\$44,711	20.7%	2.2%	11.0%
York	South Carolina	14.5	293,673	\$89,095	8.4%	2.2%	3.7%
Orangeburg	South Carolina	13.8	83,253	\$45,675	22.4%	3.1%	9.0%
Chester	South Carolina	13.8	32,182	\$52,458	18.7%	2.5%	6.3%
Holmes	Mississippi	13.5	16,191	\$32,538	36.4%	1.9%	15.9%
Lincoln	North Carolina	12.7	92,716	\$80,016	11.2%	2.9%	3.3%
Lake	Tennessee	12.5	6,579	\$28,814	32.2%	2.2%	18.3%
Alexander	North Carolina	12.2	36,412	\$65,354	12.6%	2.4%	3.1%
Quitman	Mississippi	11.0	5,774	\$32,412	29.6%	2.4%	10.9%
Alexander	Illinois	10.7	4,875	\$47,043	19.0%	4.4%	10.1%
Gaston	North Carolina	10.7	234,881	\$67,478	13.5%	2.3%	5.1%
Osage	Missouri	10.6	13,402	\$76,681	8.6%	2.6%	5.6%

Table 2: There are a few counties with high population density change but low total population.
All demographic information comes from the American Community Survey in 2022.

Demographic Profile

Counties with Most Decreased Population Density on 2026-02-02

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-12.5	99,652	\$62,581	13.5%	2.7%	4.1%
Carroll	Ohio	-9.6	26,659	\$64,835	12.9%	3.5%	6.8%
Jefferson	Arkansas	-8.9	64,802	\$51,096	18.6%	2.8%	7.5%
Clay	West Virginia	-6.7	7,835	\$42,318	26.4%	3.5%	10.3%
Fannin	Georgia	-6.7	25,742	\$57,073	13.7%	2.9%	3.0%
Taliaferro	Georgia	-6.1	1,558	\$52,500	20.4%	2.9%	5.8%
Brown	Indiana	-5.4	15,606	\$78,528	7.8%	4.5%	3.5%
Swain	North Carolina	-5.4	14,013	\$52,349	19.1%	3.8%	6.6%
Taylor	Georgia	-5.2	7,786	\$41,788	32.1%	4.2%	15.5%
Greene	Alabama	-4.8	7,424	\$29,200	40.1%	4.5%	13.3%
Watauga	North Carolina	-4.5	55,123	\$51,693	25.5%	2.3%	4.4%
White	Georgia	-3.6	28,810	\$70,666	12.4%	3.2%	2.6%
Gilmer	Georgia	-3.4	32,426	\$74,499	16.0%	3.2%	2.9%
Tishomingo	Mississippi	-3.3	18,660	\$53,040	18.0%	2.7%	4.8%
Morgan	Indiana	-3.3	72,659	\$79,429	9.3%	3.2%	3.6%

Table 3: Most of the counties with the lowest population density change have fairly high vehicular access. All demographic information comes from the American Community Survey in 2022.

Counties with Most Increased Population Density on 2026-02-12

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Issaquena	Mississippi	59.7	928	\$31,429	16.2%	2.7%	14.0%
Fairfield	South Carolina	45.2	20,550	\$47,885	15.6%	3.2%	5.9%
Osage	Missouri	12.6	13,402	\$76,681	8.6%	2.6%	5.6%
Perry	Alabama	12.6	8,031	\$37,654	31.4%	3.3%	8.0%
New Madrid	Missouri	10.6	15,731	\$51,881	17.7%	3.3%	7.2%
Norton	Virginia	10.1	3,577	\$41,495	26.3%	2.3%	10.7%
Brown	Illinois	9.9	6,322	\$67,917	9.2%	3.1%	8.5%
Humphreys	Mississippi	9.5	7,395	\$33,731	27.0%	3.1%	13.0%
Coahoma	Mississippi	9.3	20,540	\$37,461	35.5%	2.3%	14.9%
East Carroll	Louisiana	8.9	7,072	\$33,341	35.2%	1.3%	9.2%
Sumter	Alabama	8.4	11,878	\$33,310	26.5%	2.7%	9.5%
Leflore	Mississippi	7.9	27,141	\$35,277	31.5%	2.1%	14.7%
Holmes	Mississippi	7.8	16,191	\$32,538	36.4%	1.9%	15.9%
Lee	Arkansas	7.6	8,359	\$34,375	39.4%	2.3%	16.7%
Reynolds	Missouri	7.4	6,010	\$45,028	17.3%	3.8%	7.4%

Table 4: Issaquena has the highest population density but notably small population size. All demographic information comes from the American Community Survey in 2022.

Counties with Most Decreased Population Density on 2026-02-12

Source: Bing Tiles

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Sevier	Tennessee	-19.6	99,652	\$62,581	13.5%	2.7%	4.1%
Laurens	Georgia	-15.6	49,798	\$55,010	22.6%	2.3%	6.3%
Fannin	Georgia	-13.8	25,742	\$57,073	13.7%	2.9%	3.0%
Alleghany	North Carolina	-13.8	11,174	\$47,172	16.0%	2.6%	6.0%
Clay	West Virginia	-11.3	7,835	\$42,318	26.4%	3.5%	10.3%
Gilmer	Georgia	-10.1	32,426	\$74,499	16.0%	3.2%	2.9%
Ashe	North Carolina	-9.1	26,950	\$56,866	13.5%	3.5%	4.4%
Wilkes	North Carolina	-8.7	65,935	\$51,721	16.7%	2.6%	5.8%
White	Georgia	-8.6	28,810	\$70,666	12.4%	3.2%	2.6%
Watauga	North Carolina	-8.5	55,123	\$51,693	25.5%	2.3%	4.4%
Macon	North Carolina	-8.1	37,941	\$59,061	11.6%	3.2%	6.1%
Brown	Indiana	-7.3	15,606	\$78,528	7.8%	4.5%	3.5%
Taylor	Georgia	-7.1	7,786	\$41,788	32.1%	4.2%	15.5%
Hocking	Ohio	-7.0	27,799	\$62,960	16.0%	3.3%	5.7%
Morgan	Indiana	-6.8	72,659	\$79,429	9.3%	3.2%	3.6%

Table 5: Similar to Feb 2nd, the counties with strong negative population density change also have high levels of vehicular access. All demographic information comes from the American Community Survey in 2022.

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